STUDENT REPORT

DETAILS

Name

THANGADIPALLY ANVITHA

Roll Number

20R01A6753

EXPERIMENT

Title

WEEK 4

Description

4. Demonstration of classification rule process on dataset employee.arff using id3 algorithm.

Aim: This experiment illustrates the use of j-48 classifier in weka.the sample data set used in this experiment is "employee"data available at arff format. This document assumes that appropriate data pre processing has been performed.
Steps involved in this experiment:
Step 1: We begin the experiment by loading the data (employee.arff) into weka.
Step2: Next we select the "classify" tab and click "choose" button to select the "j48"classifier.
Step3: Now we specify the various parameters. These can be specified by clicking in the text box to the right of the chose button. In this example, we accept the default values the default version does perform some pruning but does not perform error pruning.
Step4: Under the "text "options in the main panel. We select the 10-fold cross validation as our evaluation approach. Since we don't have separate evaluation data set, this is necessary to get a reasonable idea of accuracy of generated model.
Step-5: We now click "start" to generate the model .the ASCII version of the tree as well as evaluation statistic will appear in the right panel when the model construction is complete.
Step-6: Note that the classification accuracy of model is about 69%.this indicates that we may find more work. (Either in preprocessing or in selecting current parameters for the classification)
Step-7: Now weka also lets us a view a graphical version of the classification tree. This can be done by right clicking the last result set and selecting "visualize tree" from the pop-up menu.
Step-8: We will use our model to classify the new instances.
Step-9: In the main panel under "text "options click the "supplied test set" radio button and then click the "set" button. This wills pop-up a window which will allow you to open the file containing test instances.

Description: Classification is a data mining function that assigns items in a collection to target categories or classes. The goal of classification is to accurately predict the target class for each case in the data. For example, a classification model could be used to identify loan applicants as low, medium, or high credit risks.

Procedure and Overvation:

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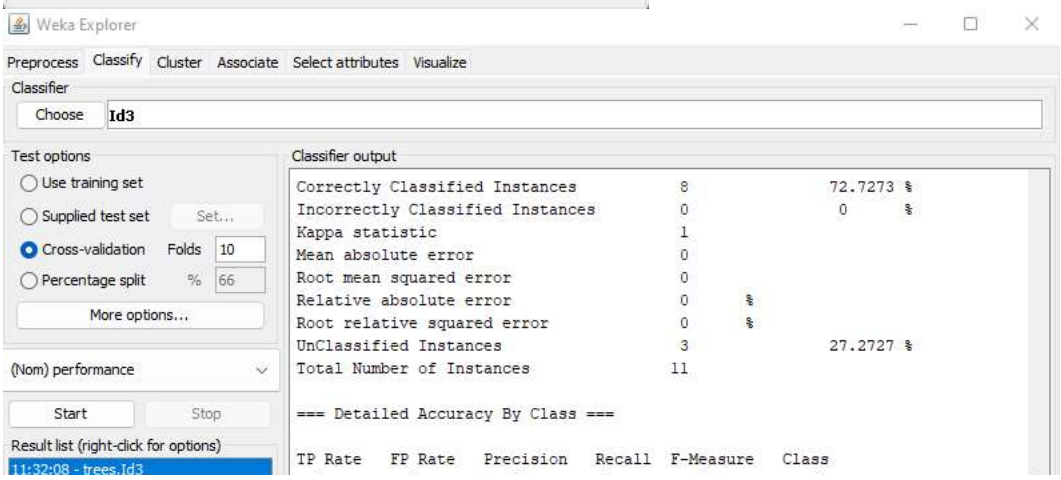
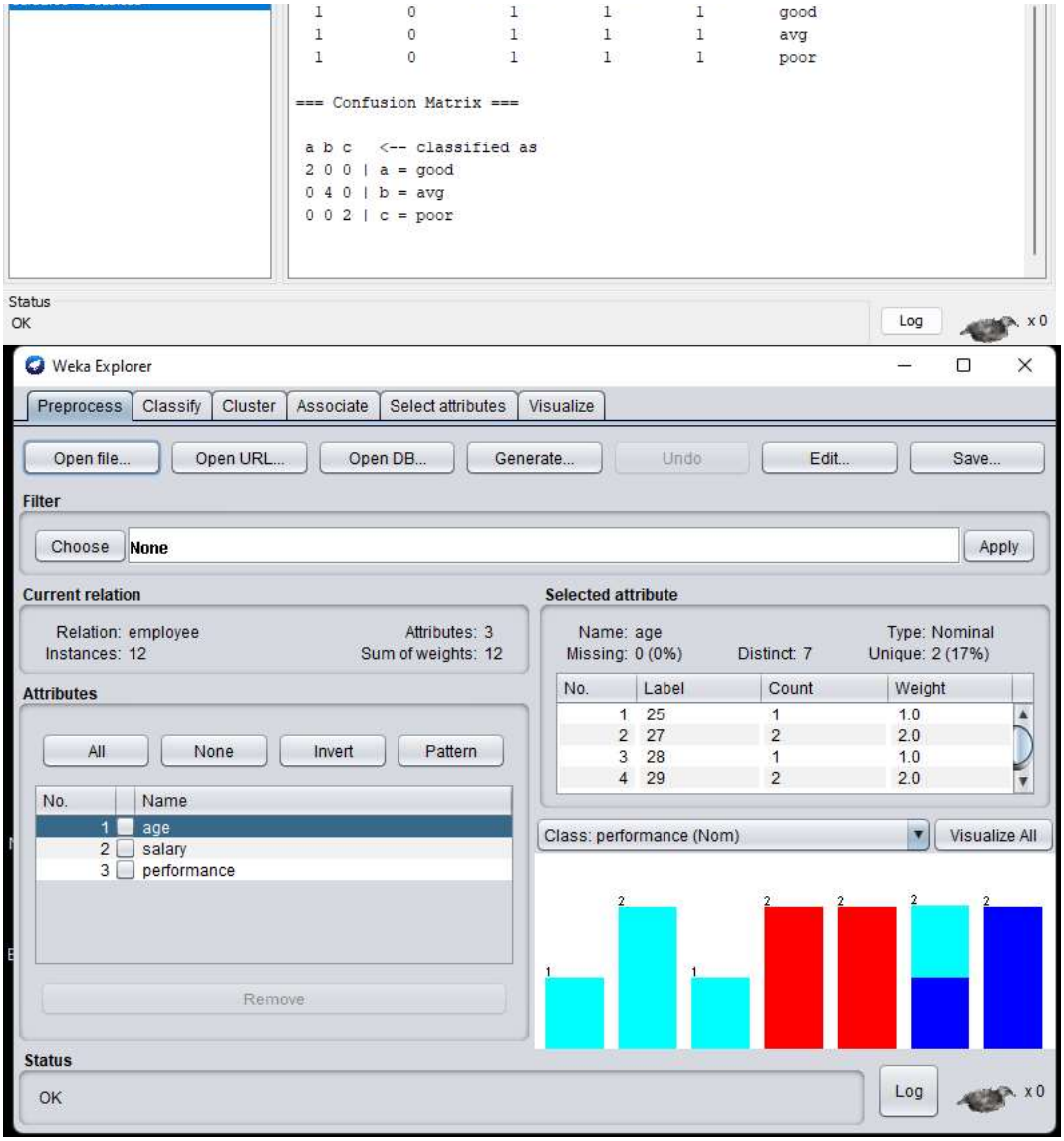
```
choose->weka-> classifier->Trees->id3->apply.

right click of id3 algorithm then visualize error.


Data set employee.arff:

@relation employee

@attribute age {25, 27, 28, 29, 30, 35, 48}

@attribute salary{10k,15k,17k,20k,25k,30k,35k,32k}

@attribute performance {good, avg, poor}

@data

%

25, 10k, poor

27, 15k, poor

27, 17k, poor

28, 17k, poor

29, 20k, avg

30, 25k, avg

29, 25k, avg

30, 20k, avg

35, 32k, good

48, 35k, good
```

48, 32k,good

%

Weka Classifier Visualize: 11:32:08 - trees.Id3 (employee)
X: Instance_number (Num)
Y: Instance_number (Num)
Colour: performance (Nom)
Select Instance
Reset
Clear
Save
Jitter
Plot: employee_predicted
10
5
0
0
5
10
B
Class colour
good avg poor


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
Id3
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) performance
Start
Stop
Result list (right-click for options)
11:32:08 - trees.Id3
Classifier output
Correctly Classified Instances 8 72.7273 %
Incorrectly Classified Instances 0 0 %
Kappa statistic 1
Mean absolute error 0
Root mean squared error 0
Relative absolute error 0 %
Root relative squared error 0 %
UnClassified Instances 3 27.2727 %
Total Number of Instances 11
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure Class



VIVA RESPONSES

Q) Use of ODS is?

Both b and c

Q) Why is the snowflake schema applied?

Normalization

Q) Identify the incorrect option related to the surrogate key.

The primary key of the fact table

Q) Identify the type of relationship between fact and dimension table in a star schema.

One to many

Q) Choose the incorrect property of the data warehouse.

Volatile